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# ============================================================
# EXPERIMENT NO. 2
# Exploratory Data Analysis (EDA) on a Real-World
# Business Dataset using Python
#
# Dataset: Netflix Movies and TV Shows
# ============================================================


# ------------------------------------------------------------
# 1. IMPORT REQUIRED LIBRARIES
# ------------------------------------------------------------

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

print("Libraries imported successfully!")
print("=" * 70)


# ------------------------------------------------------------
# 2. LOAD DATASET
# ------------------------------------------------------------

df = pd.read_csv("netflix_titles.csv")

print("Netflix dataset loaded successfully!")
print("=" * 70)


# ------------------------------------------------------------
# 3. BASIC DATASET EXPLORATION
# ------------------------------------------------------------

print("\nFIRST 5 ROWS:")
display(df.head())

print("\nDATASET SHAPE:")
print(df.shape)

print("\nCOLUMN NAMES:")
print(df.columns.tolist())

print("\nDATASET INFORMATION:")
df.info()

print("\nSTATISTICAL SUMMARY:")
display(df.describe(include="all"))


# ------------------------------------------------------------
# 4. CHECK MISSING VALUES
# ------------------------------------------------------------

print("\nMISSING VALUES:")
print(df.isnull().sum())


# ------------------------------------------------------------
# 5. CHECK DUPLICATES
# ------------------------------------------------------------

print("\nDUPLICATE RECORDS:")
print(df.duplicated().sum())


# ------------------------------------------------------------
# 6. VALUE COUNTS
# ------------------------------------------------------------

print("\nCONTENT TYPE:")
print(df["type"].value_counts())

print("\nTOP 10 COUNTRIES:")
print(df["country"].value_counts().head(10))

print("\nTOP 10 RATINGS:")
print(df["rating"].value_counts().head(10))
```

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# ============================================================
# UNIVARIATE ANALYSIS
# ============================================================

# ------------------------------------------------------------
# 7. DISTRIBUTION OF RELEASE YEAR
# ------------------------------------------------------------

plt.figure(figsize=(10, 5))

sns.histplot(
    df["release_year"],
    bins=30,
    kde=True
)

plt.title("Distribution of Release Year")
plt.xlabel("Release Year")
plt.ylabel("Number of Titles")

plt.show()


# ------------------------------------------------------------
# 8. CONTENT TYPE COUNT PLOT
# ------------------------------------------------------------

plt.figure(figsize=(7, 5))

sns.countplot(
    data=df,
    x="type"
)

plt.title("Movies vs TV Shows")
plt.xlabel("Content Type")
plt.ylabel("Number of Titles")

plt.show()


# ------------------------------------------------------------
# 9. TOP 10 COUNTRIES
# ------------------------------------------------------------

top_countries = df["country"].value_counts().head(10)

plt.figure(figsize=(10, 6))

top_countries.plot(kind="bar")

plt.title("Top 10 Countries by Number of Netflix Titles")
plt.xlabel("Country")
plt.ylabel("Number of Titles")

plt.xticks(rotation=45)

plt.show()


# ------------------------------------------------------------
# 10. TOP 10 RATINGS
# ------------------------------------------------------------

top_ratings = df["rating"].value_counts().head(10)

plt.figure(figsize=(10, 5))

top_ratings.plot(kind="bar")

plt.title("Top Netflix Content Ratings")
plt.xlabel("Rating")
plt.ylabel("Number of Titles")

plt.xticks(rotation=45)

plt.show()


# ============================================================
# BIVARIATE ANALYSIS
# ============================================================
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# ------------------------------------------------------------
# 11. MOVIES VS TV SHOWS BY RELEASE YEAR
# ------------------------------------------------------------

type_year = pd.crosstab(
    df["release_year"],
    df["type"]
)

type_year = type_year.tail(20)

type_year.plot(
    kind="bar",
    figsize=(12, 6)
)

plt.title("Movies and TV Shows by Release Year")
plt.xlabel("Release Year")
plt.ylabel("Number of Titles")

plt.xticks(rotation=45)

plt.show()


# ------------------------------------------------------------
# 12. RATING VS CONTENT TYPE
# ------------------------------------------------------------

plt.figure(figsize=(12, 6))

sns.countplot(
    data=df,
    x="rating",
    hue="type"
)

plt.title("Content Rating by Type")
plt.xlabel("Rating")
plt.ylabel("Number of Titles")

plt.xticks(rotation=45)

plt.show()


# ============================================================
# MULTIVARIATE ANALYSIS
# ============================================================

# ------------------------------------------------------------
# 13. CREATE NUMERICAL FEATURES
# ------------------------------------------------------------

# Extract first number from duration
df["duration_num"] = pd.to_numeric(
    df["duration"].str.extract(r"(\d+)")[0],
    errors="coerce"
)

# Encode content type
df["type_encoded"] = df["type"].map({
    "Movie": 0,
    "TV Show": 1
})

# Calculate number of days since release
df["content_age"] = 2025 - df["release_year"]


# ------------------------------------------------------------
# 14. CORRELATION MATRIX
# ------------------------------------------------------------

numeric_df = df[
    [
        "release_year",
        "duration_num",
        "type_encoded",
        "content_age"
    ]
].copy()
```

```
correlation = numeric_df.corr()

print("\nCORRELATION MATRIX:")
display(correlation)


# ------------------------------------------------------------
# 15. CORRELATION HEATMAP
# ------------------------------------------------------------

plt.figure(figsize=(8, 6))

sns.heatmap(
    correlation,
    annot=True,
    cmap="coolwarm",
    fmt=".2f"
)

plt.title("Correlation Heatmap")

plt.show()


# ------------------------------------------------------------
# 16. BOXPLOT
# ------------------------------------------------------------

plt.figure(figsize=(8, 5))

sns.boxplot(
    data=df,
    x="type",
    y="duration_num"
)

plt.title("Duration Distribution by Content Type")
plt.xlabel("Content Type")
plt.ylabel("Duration")

plt.show()


# ------------------------------------------------------------
# 17. SCATTER PLOT
# ------------------------------------------------------------

plt.figure(figsize=(9, 6))

sns.scatterplot(
    data=df,
    x="release_year",
    y="duration_num",
    hue="type"
)

plt.title("Release Year vs Duration")
plt.xlabel("Release Year")
plt.ylabel("Duration")

plt.show()


# ============================================================
# ADDITIONAL BUSINESS INSIGHTS
# ============================================================

# ------------------------------------------------------------
# 18. YEAR WITH MOST CONTENT
# ------------------------------------------------------------

year_counts = df["release_year"].value_counts()

print("\nYEAR WITH MOST CONTENT:")
print(year_counts.head(10))


# ------------------------------------------------------------
# 19. MOVIE / TV SHOW TOTALS
# ------------------------------------------------------------

print("\nTOTAL MOVIES AND TV SHOWS:")
print(df["type"].value_counts())
```

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# ------------------------------------------------------------
# 20. MOST COMMON GENRES
# ------------------------------------------------------------

genres = (
    df["listed_in"]
    .dropna()
    .str.split(", ")
    .explode()
)

print("\nTOP 10 GENRES:")
print(genres.value_counts().head(10))


# ------------------------------------------------------------
# 21. TOP DIRECTORS
# ------------------------------------------------------------

print("\nTOP 10 DIRECTORS:")
print(df["director"].value_counts().head(10))


# ------------------------------------------------------------
# 22. FINAL SUMMARY
# ------------------------------------------------------------

print("\n" + "=" * 70)
print("EDA SUMMARY")
print("=" * 70)

print("Total Rows:", df.shape[0])
print("Total Columns:", df.shape[1])

print("\nMost Common Content Type:")
print(df["type"].mode()[0])

print("\nMost Common Rating:")
print(df["rating"].mode()[0])

print("\nMost Common Genre:")
print(genres.mode()[0])

print("\nMost Common Country:")
print(df["country"].dropna().str.split(", ").explode().mode()[0])


# ============================================================
# SAVE EDA DATASET
# ============================================================

df.to_csv(
    "Netflix_EDA_Analysis.csv",
    index=False
)

print("\nEDA dataset saved successfully!")
print("File: Netflix_EDA_Analysis.csv")


# ============================================================
# QUESTIONS AND ANSWERS
# ============================================================


# Q1. What is Exploratory Data Analysis (EDA), and why is it
# performed before machine learning?
#
# Answer:
# EDA is the process of examining and visualizing a dataset
# to understand its structure, patterns, relationships and
# anomalies. It is performed before machine learning to identify
# data quality problems, understand important features and
# select suitable variables for model building.


# Q2. Differentiate between univariate, bivariate, and
# multivariate analysis with suitable examples.
#
# Answer:
#
# Univariate Analysis:
```

```
# Analysis of one variable.
# Example: Distribution of Netflix release years.
#
# Bivariate Analysis:
# Analysis of two variables.
# Example: Release year vs content type.
#
# Multivariate Analysis:
# Analysis of more than two variables.
# Example: Release year, duration and content type together.


# Q3. What insights can be obtained from a correlation heatmap?
#
# Answer:
# A correlation heatmap shows the strength and direction of
# relationships between numerical variables. Values close to
# +1 indicate strong positive correlation, values close to -1
# indicate strong negative correlation, and values near 0
# indicate weak or no linear relationship.


# Q4. Explain the purpose of histograms, box plots, and
# scatter plots in EDA.
#
# Answer:
#
# Histogram:
# Shows the distribution and frequency of numerical data.
#
# Box Plot:
# Shows median, spread and possible outliers.
#
# Scatter Plot:
# Shows the relationship between two numerical variables.


# Q5. How can EDA help identify data quality issues
# before analysis?
#
# Answer:
# EDA can identify missing values, duplicate records, outliers,
# incorrect data types, inconsistent categories and unusual
# distributions. This helps clean the dataset before further
# analysis or machine learning.


# Q6. Why is correlation important in predictive analytics?
# Can correlation imply causation?
#
# Answer:
# Correlation helps identify relationships between variables
# and can help in feature selection. However, correlation does
# not imply causation. Two variables may be correlated without
# one causing the other.


# Q7. Which visualization would you use to analyze categorical
# and numerical variables? Justify your choice.
#
# Answer:
#
# Categorical Variables:
# Bar charts and count plots are suitable because they show
# the frequency of different categories.
#
# Numerical Variables:
# Histograms, box plots and scatter plots are useful because
# they show distribution, outliers and relationships.


# Q8. What business insights can be derived from the Netflix
# dataset through EDA?
#
# Answer:
# EDA can identify:
# 1. The number of Movies and TV Shows.
# 2. Countries producing the most content.
# 3. Most common content ratings.
# 4. Popular genres.
# 5. Trends in content release over the years.
# These insights can support content planning and business
# decision-making.
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# Q9. How does EDA contribute to feature selection
# and model building?
#
# Answer:
# EDA helps identify important variables, relationships,
# redundant features, outliers and unusual patterns.
# This helps analysts select useful features and prepare
# better inputs for machine learning models.


# Q10. What challenges might arise while performing EDA
# on large-scale real-world datasets?
#
# Answer:
# Major challenges include:
# 1. Large data size.
# 2. High processing and memory requirements.
# 3. Missing and inconsistent data.
# 4. Large number of features.
# 5. Slow visualization and computation.
# 6. Difficulty in identifying meaningful patterns.


# ============================================================
# END OF EXPERIMENT 2
# ============================================================
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```
Libraries imported successfully!
======================================================================
Netflix dataset loaded successfully!
======================================================================

FIRST 5 ROWS:
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	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020	PG-13	90 min	Documentaries	As her father nears the end of his life, filmm...
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	September 24, 2021	2021	TV-MA	2 Seasons	International TV Shows, TV Dramas, TV Mysteries	After crossing paths at a party, a Cape Town t...
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	September 24, 2021	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Act...	To protect his family from a powerful drug lor...
3	s4	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	September 24, 2021	2021	TV-MA	1 Season	Docuseries, Reality TV	Feuds, flirtations and toilet talk go down amo...
4	s5	TV Show	Kota Factory	NaN	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	September 24, 2021	2021	TV-MA	2 Seasons	International TV Shows, Romantic TV Shows, TV ...	In a city of coaching centers known to train I...

```
DATASET SHAPE:
(8807, 12)

COLUMN NAMES:
['show_id', 'type', 'title', 'director', 'cast', 'country', 'date_added', 'release_year', 'rating', 'duration', 'listed_in', 'description']

DATASET INFORMATION:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 12 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   show_id       8807 non-null   object
 1   type          8807 non-null   object
 2   title         8807 non-null   object
 3   director      6173 non-null   object
 4   cast          7982 non-null   object
 5   country       7976 non-null   object
 6   date_added    8797 non-null   object
 7   release_year  8807 non-null   int64
 8   rating        8803 non-null   object
 9   duration      8804 non-null   object
 10  listed_in     8807 non-null   object
 11  description   8807 non-null   object
dtypes: int64(1), object(11)
memory usage: 825.8+ KB
```

STATISTICAL SUMMARY:

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description
count	8807	8807	8807	6173	7982	7976	8797	8807.000000	8803	8804	8807	8807
unique	8807	2	8807	4528	7692	748	1767	NaN	17	220	514	8775
top	s8807	Movie	Zubaan	Rajiv Chilaka	David Attenborough	United States	January 1, 2020	NaN	TV-MA	1 Season	Dramas, International Movies	Paranormal activity at a lush, abandoned prope...
freq	1	6131	1	19	19	2818	109	NaN	3207	1793	362	4
mean	NaN	NaN	NaN	NaN	NaN	NaN	NaN	2014.180198	NaN	NaN	NaN	NaN
std	NaN	NaN	NaN	NaN	NaN	NaN	NaN	8.819312	NaN	NaN	NaN	NaN
min	NaN	NaN	NaN	NaN	NaN	NaN	NaN	1925.000000	NaN	NaN	NaN	NaN
25%	NaN	NaN	NaN	NaN	NaN	NaN	NaN	2013.000000	NaN	NaN	NaN	NaN
50%	NaN	NaN	NaN	NaN	NaN	NaN	NaN	2017.000000	NaN	NaN	NaN	NaN